THE SMAKS PROJECT · KOCHI



Moonlight

Old medicines, new jobs. An AI explorer for drug-repurposing clues, built on open data.

Startup Buildathon India 2026 · Google DeepMind × Google Cloud

01 · THE PROBLEM

The evidence for repurposing old medicines already exists. Almost nobody can read it.

Repurposing is the fast lane

An approved medicine has passed safety trials. A new use can be studied in a fraction of the time and cost of inventing a new drug.

Rare diseases need it most

Thousands of rare diseases get little research funding. Patient groups hunt for leads with no bioinformatics staff.

The data is locked

Open Targets links drugs, proteins and diseases, but behind GraphQL, ontology IDs and dense score tables.

A medicine built for one disease may grab a protein that belongs to another.



Medicine

approved, already tested in people

Protein

the knob the medicine turns up or down

Disease

linked to that protein in the research record

The technique is called knowledge-graph drug repurposing. Moonlight finds the clues; scientists test them.

03 · HOW IT WORKS

Five stages. AI at both ends, verified data in the middle.

1 · TRANSLATE

Gemini

“rare muscle disease” becomes candidate disease names. Open Targets search confirms the real ID.

2 · FETCH

Python + GraphQL

Disease to associated targets with scores; targets to drugs with mechanism, action type and phase.

3 · FILTER + RANK

Our graph engine

Drop drugs already linked to the disease. Rank the rest by proteins hit, weighted by association score.

4 · EXPLAIN

Gemini

Each path’s records become plain language. Input is only the retrieved data. No invented facts.

5 · MAP

Streamlit

Click a medicine and its path lights up: medicine to protein to disease, with source and evidence strength.

Dark stages are AI. Light stages are deterministic code and verified data.

04 · TECH STACK

Open data, one model, a few hundred lines of Python.

Layer	Choice	Why
Data	Open Targets Platform GraphQL API	Free, open-licensed, maintained by EMBL-EBI and Sanger. Disease, target and drug records with evidence scores.
Backup data	ChEMBL	Open bioactivity and mechanism data if a field is missing upstream.
Engine	Python 3, requests, in-memory graph	Two-hop walk, known-drug filter, deterministic ranking. Testable, no black box.
AI	Gemini API (Flash for translate and explain; Pro for reports)	Fast and cheap at the edges. Grounded prompts carry only retrieved records.
Interface	Streamlit + interactive graph component	Search, ranked list, clickable path map, permanent safety banner.
Hosting	Streamlit Community Cloud, later Google Cloud Run	Free public link on day one; a path to scale with Google credits.
Build process	GitHub + agentic coding (Claude Code), small verified steps	Every step run, inspected and committed. Schema introspected live, never assumed.

Secrets live in environment secrets, never in code. Estimated v1 to v2 build: 7 to 11 hands-on hours.

A lookup shows what is known. A detective removes it.

Open Targets already lists the drugs being tested for a disease. Those are not clues. Moonlight subtracts them, then ranks what remains by how many disease-linked proteins each medicine touches and how strong those links are.

```
candidates = drugs(targets(disease))    - known(disease)    → rank(score · hits)
```

06 · SAFETY BY DESIGN

Clues, never answers. The guardrails were written before the first line of code.

- Every page: for research exploration, not medical advice. Never change a medicine without a doctor.
- Every result shows its source and evidence strength. No AI-invented links; Gemini explains only retrieved records.
- Every result shows direction: does the medicine turn the protein up or down? The wrong direction can worsen a disease.
- Honest gaps are stated: when the data does not say which direction the disease needs, the result says so.

Framed as “connections worth reading about”, never as treatments.

07 · USERS AND BUSINESS MODEL

Free for the curious. Paid for the organised.

FREE TIER

Students, teachers, patient families, curious adults

- Plain-language search and clickable map
- Sources, evidence strength and direction on every result
- Builds trust and reach; the underlying data is public anyway

ORGANISATION TIER · ANNUAL SUBSCRIPTION

Rare-disease foundations, advocacy groups, research teams, biotech analysts

- Saved disease watchlists and alerts on each database release
- Team workspaces
- Exportable, citation-backed briefing reports for researchers and grant panels
- Sponsored access for small patient groups

Primary market: India. Secondary: US and UK, home to the largest rare-disease foundation ecosystems.

We complement the research platforms. We do not compete with the scientists.

Alternative	Built for	Moonlight's difference
Open Targets Platform website	Scientists who know the ontology and the scores	Plain-language search, a map, and the known-drug filter on top of their data
AI repurposing programmes (e.g. Every Cure)	Researchers and clinicians, via large internal pipelines	Self-serve, public, explainable tool for non-specialists and advocacy groups
Academic knowledge-graph tools	Users with graph and query skills	No setup, no code; sources, direction and evidence strength on every result

Non-specialist first. Explanations a curious 12-year-old can follow.

Detective, not lookup. Known drugs removed, remaining clues ranked.

Safety first. AI explains retrieved data only; nothing invented.

The SMAKS Project: a family team from Kochi, two co-founders.

Shreshth Dugar

BUSINESS LEAD · AGENT OPERATOR

CMO of a 100-year-old listed tea plantation company and COO of a bulk tea supplier; NYU-trained journalist. Runs the coding agents, infrastructure, deployment and the business model.

Kashish Dugar, 9

PRODUCT · EXPLANATIONS · DEMO

Codes 2.5 hours a day and blogs it. Finished Grokking Algorithms in 18 days: hash tables, BFS, Dijkstra, dynamic programming and k-nearest neighbours, each built from scratch in Python. Deployed a Django app. Started Rust in late October and is already rebuilding programs from memory. Designs the ranking on paper, judges every explanation, presents the demo.

Stage: ideation

BUILDATHON PLAN

- v1 terminal engine: 2 to 4 hours
- Plumbing test: sildenafil and PDE5 appear with the filter off
- v2 Streamlit map on a public link: 3 to 5 hours
- Google credits fund Gemini usage and Cloud Run

Blog: kashishdugar2017.github.io · Repo: github.com/kashishdugar2017/moonlight · Contact: shreshthdugar@gmail.com